means that they were owned by the original equipment manufacturers (OEMs). They wanted to run an operating system on multiple pieces of hardware, which was not possible at all.

Some of the hyperscalers like Facebook and Google found an opportunity and started using some of the white box hardware. White box hardware can be commercially used right off the shelf. These are servers which are based on open hardware designs that you can easily procure.

After you have taken the white box hardware you can then basically put any open software in there. Now, open software can be taken from various vendors and can still run on the white box made by any of the multiple companies. You can even change the software later, if you need to, and still be able to run it on the hardware. This is also called disaggregation.

The movement from proprietary solutions to open and disaggregated software and hardware in the data centres started in 2012 (see Figure 1).

The white box use was very miniscule at about 2% in 2012. But by 2019 the white box use dominated the market and became around 59%. This was an eye opener for the telecom industry and a lot of enterprises too. Since then, disaggregation and openness has become an effective standard in the data centre market.

This was also the time when our company (Niral Networks) started to envisage what it wanted to do. Interestingly, even big companies like Cisco are moving towards disaggregation, because that is where the market is headed.

Transformation towards openness and disaggregation in 5G RAN and core

Initially, the radio access network (RAN) was proprietary but with the introduction of open radio access network (O-RAN) it has been divided into three parts now. The first is the radio unit (RU) which acts like antennas. Then there is a distributed unit (DU) and a centralised unit (CU). Now, instead of making use of the hardware you can simply use the cloud to run DU and CU.

The important parts in the 5G are the radio network, as mentioned previously, and the transport network that sends the radio signals to the centralised core. There is a transport network between the 5G network and the centralised cloud using a router which is also getting disaggregated.

NiralOS network operating system

- Private 5G Core Cloud is a native private 5G core software for mobility, authentication, security, session, and policy management. It contains the 5G network functions like AMF, SMF, AUSF, DM, NRF, and UPF. Niral 5G core also has a compact user plane function (UPF) to provide local breakout within an enterprise when integrated with TSP's centralised 5G core.
- Mobile edge cloud Kubernetes and virtualised edge cloud infrastructure creates a mobile edge cloud within an enterprise with open APIs to host third-party applications like AR/VR, robotics, drones, AI/ML, and video analytics for low latency and privacy.
- Centralised controller provides centralised management, orchestration, zero touch provisioning, and monitoring of multiple private 5G networks and mobile edge cloud at various sites. The controller can be hosted in the public cloud to centrally manage and monitor multiple private 5G networks.

NiralOS specifications

- Release-16 compliant 5G core for private 5G deployment
- 5G network functions like UPF, AMF. SMF, AUSF, UDM, and NRF
- Kubernetes based cloud-native network function
- DPDK+VPP based user plane (UPF) acceleration - linear scaling per CPU core
- UPF local breakout for easy integration to TSP's centralised 5G core
- Support of N1, N2, N3, N4, and service based interfaces for 5G SBA
- 5G core deployed on COTS hardware of various form factors and integrated with 5G radio
- Kubernetes and virtualised cloud agnostic edge cloud to host third-party applications
- Open APIs for integration of third-party applications to Niral 5G core and edge platform
- Web based dashboard for subscriber provisioning configuration and management

We have done some indigenous end-to-end integration with a radio company in India, so we can provide an end-to-end 5G system. We have recently been awarded for innovation in 5G and look forward to achieving more such feats. END

This article is based on a tech talk by Abhijit Chaudhary of Niral Networks at Open Source Conference 2022 that was organised by Samsung R&D Institute India, Bengaluru, and IEEE ComSoc Bengaluru Chapter. It was transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research.

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